Azure DevOps Pipeline – Employee Attendance Tracker

Objective
The Azure DevOps pipeline automates the execution of the Employee Attendance and Productivity Tracker project. The pipeline runs weekly, processes attendance data, performs analytics using Python and PySpark, and generates reports automatically.

Pipeline Workflow
- Start the Employee Attendance Pipeline
- Run employee_attendance_analysis.py for attendance analytics
- Run employee_attendance_pyspark.py for PySpark-based processing
- Generate attendance and productivity reports
- Store processed output files
- Publish reports automatically

Pipeline Script

```
import os
print("Starting Employee Attendance Pipeline")
os.system("python employee_attendance_analysis.py")
os.system("python employee_attendance_pyspark.py")
print("Employee Attendance Pipeline Executed Successfully")
```

YAML Pipeline Configuration

```
trigger:
- main

schedules:
- cron: "0 9 * * 1"
  displayName: Weekly Monday Run
```

```
  branches:
    include:
    - main
  always: true

pool:
  vmImage: ubuntu-latest

steps:

- task: UsePythonVersion@0
  inputs:
    versionSpec: '3.x'

- script: |
    pip install pandas numpy pyspark
  displayName: Install Dependencies

- script: |
    python attendance_pipeline.py
  displayName: Run Attendance Pipeline

- task: PublishBuildArtifacts@1
  inputs:
    PathtoPublish: employee_metrics_csv
    ArtifactName: AttendanceReports
```

Azure Databricks

```
print("Final Department Productivity Report")
dept_metrics.show()
top_performers = df.orderBy(df["productivity_score"].desc())
print("Top Performers")
```

```
top_performers.select("employee_id","employee_name","department","wor
k_hours","productivity_score").show()
frequent_absentees = df.filter(df["work_hours"] < 6)
print("Frequent Absentees")
frequent_absentees.select("employee_id","employee_name",“department","
work_hours").show()

dept_metrics.write.mode("overwrite").option("header",True).csv("employee
_metrics_csv")

print("Employee Metrics CSV File Saved Successfully")
```

Final Outcome

The pipeline automatically processes employee attendance and productivity data every week and generates KPI reports for HR analysis.